

Optimistic Initial Values

3. $Q_1(a)$, the initial action-value estimates, must be picked by the user, if only to set them all to zero.

1. sample-average method

$$\therefore Q_{n+1} = Q_n + \frac{1}{n}(R_n - Q_n), \quad n \geq 1$$

$$\therefore Q_2 = R_1, \quad \forall Q_1$$

4. Optimistic initial values

$\therefore$ The bias disappears once all actions have been selected at least once.

2. weighted average method

$$\therefore Q_{n+1} = Q_n + \alpha(R_n - Q_n), \quad \alpha \in (0, 1]$$

$$= \alpha R_n + (1 - \alpha)Q_n$$

$$= (1 - \alpha)^n Q_1 + \sum_{i=1}^n (1 - \alpha)^{n-i} \alpha R_i$$

can be quite effective on stationary problems, but it is not well suited to nonstationary problems.

This criticism applies to

both sample-average methods

and weighted average methods.

$\therefore$ The bias is permanent, though decreasing over time.